

Banking Credit Risk Analysis

Berka Retail Bank, a B2C retail bank based on the public Czech banking dataset

INDUSTRY & COMPANY

Berka Retail Bank is a retail bank offering accounts, transactions, loans, and cards. We chose banking because it is rich in data and credit risk is very relevant for a data analyst.

BUSINESS MODEL: B2C

The bank earns mainly from lending interest, plus fees on cards and transactions, serving individual customers. Key processes: open account, transact, apply for a loan, then track repayment.

DATA SCALE & RANGE

1993 to 1998, six full years. Covers 5,369 clients, 4,500 accounts, ~1.05M transactions, 682 loans, and 892 cards.

BUSINESS FLOW: WHERE DATA IS GENERATED

New client	Open account	Transactions	Loan application	Repayment & status
client	account	trans	loan	loan.status

ROLE OF DATA IN DECISIONS

The bank uses transaction behavior, account activity, and loan status to watch credit risk, guide lending decisions, and group its customers.

DATASET SOURCE

Available on Kaggle as “The Berka Dataset”, a Czech banking dataset from PKDD’99.
[kaggle.com/datasets/marceloventura/the-berka-dataset](https://www.kaggle.com/datasets/marceloventura/the-berka-dataset)

Data Validation & Problem Framing

DATASET STRUCTURE & GRANULARITY

client, one row per customer

key columns: client_id, district_id

account, one row per account

key columns: account_id, frequency, date

disp, links a client to an account

key columns: client_id, account_id, type

trans, one row per transaction

key columns: amount, balance, type

loan, one row per loan

key columns: amount, duration, status

card, one row per issued card

key columns: type, issued

KPIs COMPUTABLE FROM THE DATASET

KPI	Formula	Value
Problematic Loan Rate	$(B + D) \div \text{total loans}$	11.1%
Transaction Volume	<code>COUNT(trans_id)</code>	1.05M
Avg Transaction Value	<code>AVG(trans.amount)</code>	Kč 5,925 Koruna Ceko

Also available: Card penetration $\approx$ 20% · Problematic loans = 76 of 682

OUT OF SCOPE FOR THIS DATA

✗ **How can the bank increase its revenue or profit?**

The data has no revenue, fees, operating costs, or profit figures.

✗ **Why are customers dissatisfied with the bank?**

No satisfaction scores, complaints, reviews, or feedback exist.

CORE BUSINESS QUESTION

What customer and transaction characteristics are associated with problematic loan outcomes?

Specific · relevant to credit risk · answerable by joining client + account + trans + loan on account_id.

DARCI, Scope & Problem Statement

	Role	Who
D	Decider	Head of Credit Risk
A	Accountable	Credit Risk Manager
R	Responsible	Data Analyst (me)
C	Consulted	Loan Officers, Collections Team
I	Informed	Senior Management, Branch Heads

Who decides, who owns it, and who needs to hear about it.

PROBLEM STATEMENT (SMART)

How can the credit team reduce Berka's problematic loan rate from 11% to below 5% within the next 12 months by identifying risk patterns in the 1993–1998 transaction and account data?

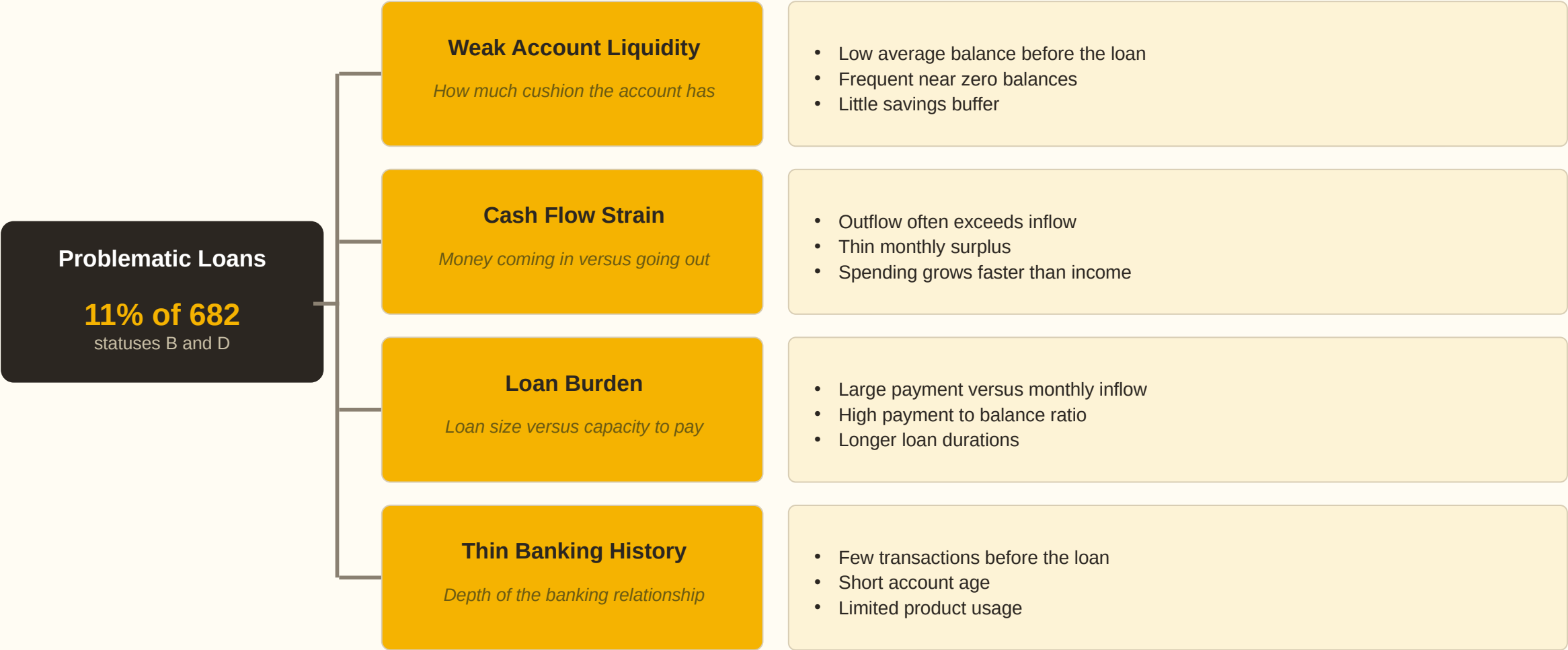
IN SCOPE

- **Transaction behavior from trans:** balance, money in vs. money out, and transaction volume
- **Account and loan activity:** account activity and loan terms from account and loan tables
- **Relationship between financial behavior and loan status:** good loans (A, C) vs. problematic loans (B, D)

OUT OF SCOPE

- **Bank revenue, fees, and profit:** no such columns exist
- **Customer satisfaction and complaints:** not recorded
- **Marketing and acquisition channels:** not in the data
- **Age or gender profiling:** only district is available

Root Cause Analysis (Issue Tree)



These four root-cause categories are mutually exclusive and collectively exhaustive for the risk factors observable in the available dataset. Each category is measurable, while problematic loan status is treated as the outcome, not a cause.

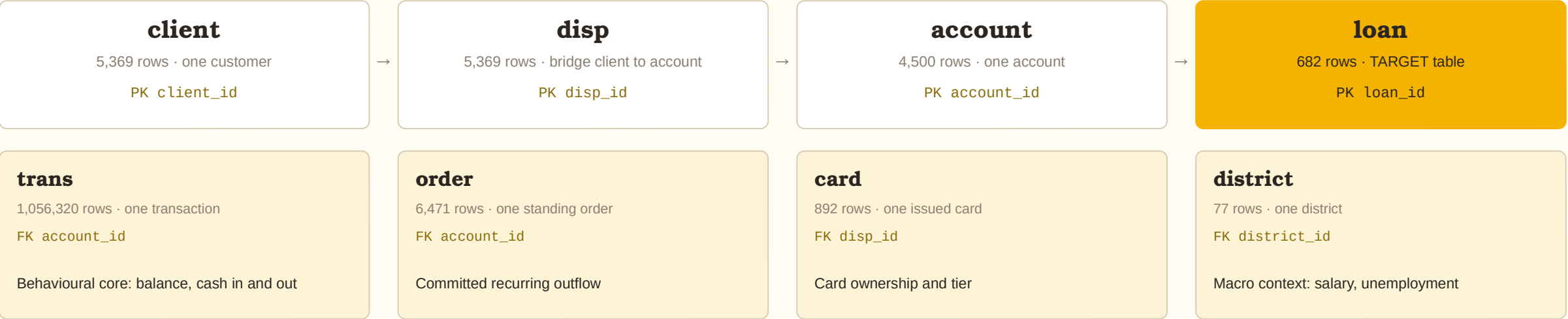
Hypotheses, Metrics & Alignment

Overall KPI: Problematic Loan Rate = $(B + D) \div \text{total loans} \approx 11\%$. Each root cause maps to a testable hypothesis and a metric from the Berka tables.

Root Cause	Hypothesis (if, then)	Key Metric	Why it matters
Weak Account Liquidity	If average balance before the loan is low, then the loan is more likely to become problematic.	Avg pre-loan balance = AVG(trans.balance before loan date)	A small cushion means less room to absorb repayments.
Cash Flow Strain	If the outflow to inflow ratio is high, then default risk rises.	Outflow to inflow = money out ÷ money in (trans)	Spending above income signals repayment stress.
Loan Burden	If the monthly payment is large versus inflow, then the loan is more likely to fail.	Loan to inflow = loan.payments ÷ avg monthly inflow	Oversized installments strain the account.
Thin Banking History	If there are few transactions before the loan, then risk is higher.	Pre-loan activity = COUNT(trans before loan date)	A short history gives the bank less signal to judge risk.

Eight tables, one spine: every loan hangs off account_id

1,056,320 transactions and 682 loans across six years, 1993 to 1998, with zero orphan rows in all eight foreign keys.



WHAT THE STRUCTURE FORCES US TO DO

Clients never touch accounts directly. The link runs through `disp`, which holds 5,369 rows for 4,500 accounts because 869 accounts carry a second user (DISPONENT). Joining loans to clients without filtering `disp` to OWNER would multiply the 682 loans and double-count risky customers, so the join is constrained and the row count asserted at 682 afterwards.

Granularity decides the analysis unit. Only 682 of 4,500 accounts ever borrowed, so every table is collapsed to one row per loan. Loan status mixes two dimensions, finished (A, B) versus running (C, D) and healthy (A, C) versus problematic (B, D), which is why the KPI is defined as $(B + D) \div \text{total}$ rather than a default rate.

Cleaning was driven by the analysis, not by a checklist

Every action answers three questions: what is the issue, why does it matter, what decision was taken.

DATA TYPES

Issue : Dates stored as YYMMDD integers (930705); birth_number hides gender and date of birth.

Why it matters : Time ordering is the backbone of the analysis, and age and gender are standard risk covariates.

Decision : Converted to datetime with the century forced to 19, decoded birth_number into gender and birth_date. Without the explicit century pandas reads 11 as 2011 and ages go negative.

MISSING VALUES

Issue : Four trans columns incomplete: bank 74.1%, account 72.0%, k_symbol 50.7%, operation 17.3%. k_symbol also hid 53,433 blank strings.

Why it matters : Imputing them would be fabrication, and dropping the rows would destroy three quarters of the behavioural data.

Decision : Kept every row, imputed nothing. The blanks are structural: a cash deposit has no counterparty bank. Analysis relies on amount, balance, date and type, all fully populated.

DUPLICATES

Issue : No full-row duplicates and no repeated primary key in any of the eight tables.

Why it matters : The real duplication risk is created by the joins, not by the source files.

Decision : No rows removed. The disp join is filtered to OWNER and the loan count asserted at 682 after the merge, proving no multiplication occurred.

OUTLIERS

Issue : IQR flags 120,215 transaction amounts (11.4%) and 18 loan amounts; 2,999 transactions sit at a negative balance.

Why it matters : Removing them is the default reflex and would have been the single most damaging decision in this project.

Decision : Retained all of them. Distributions are right skewed by nature and 18 of 682 loans is 2.6% of the target. Negative balances became a feature instead.

120,215

transaction amounts flagged by IQR

2,999

transactions at a negative balance

0

rows removed or winsorised

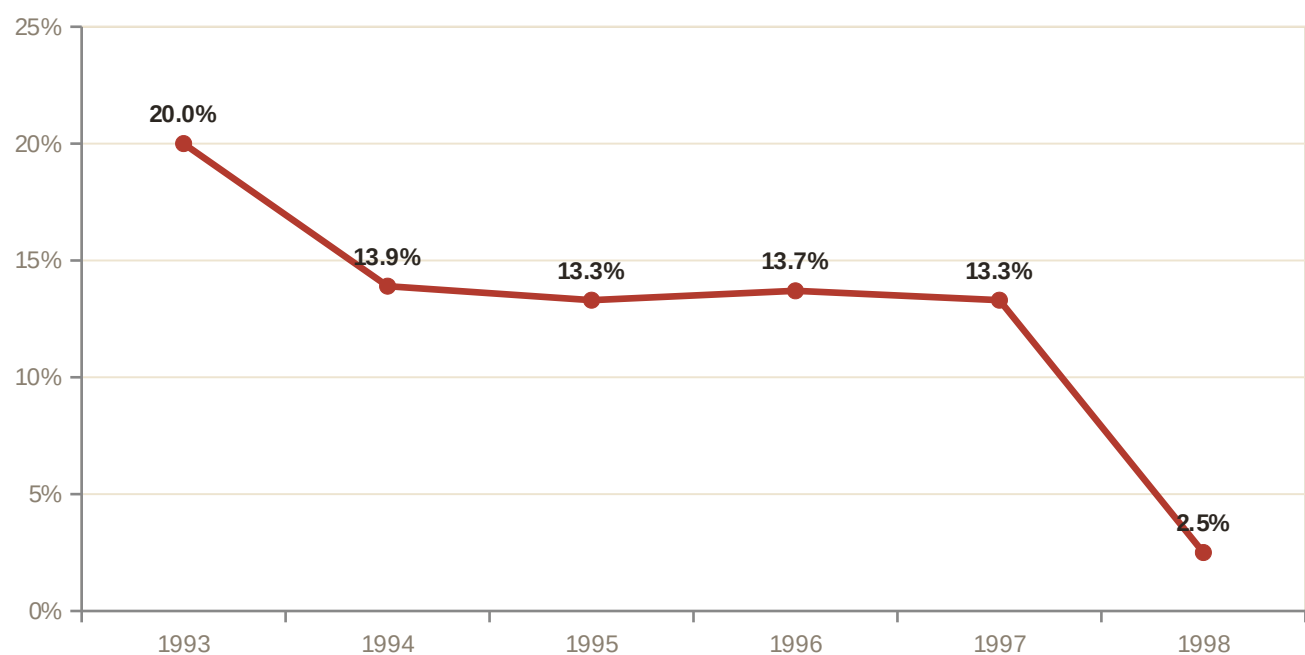
35.5%

of bad loans caught by the feature built from them

No-leakage rule: Only transactions before the loan date are used, ensuring predictors reflect information available at the time of the credit decision

The 1998 drop is maturity bias, not better underwriting

Problematic Loan Rate = $(\text{status B} + \text{D}) \div \text{total loans}$. Baseline 11.1%, 76 of 682 loans.



WHAT HAPPENED

The rate holds steady near 13 to 14% from 1994 to 1997, then collapses to 2.5% in 1998.

WHY IT HAPPENED

Loans issued in 1998 had at most twelve months to deteriorate before the data ends, while a 1994 loan had five years. Many were 60-month contracts still in their first year.

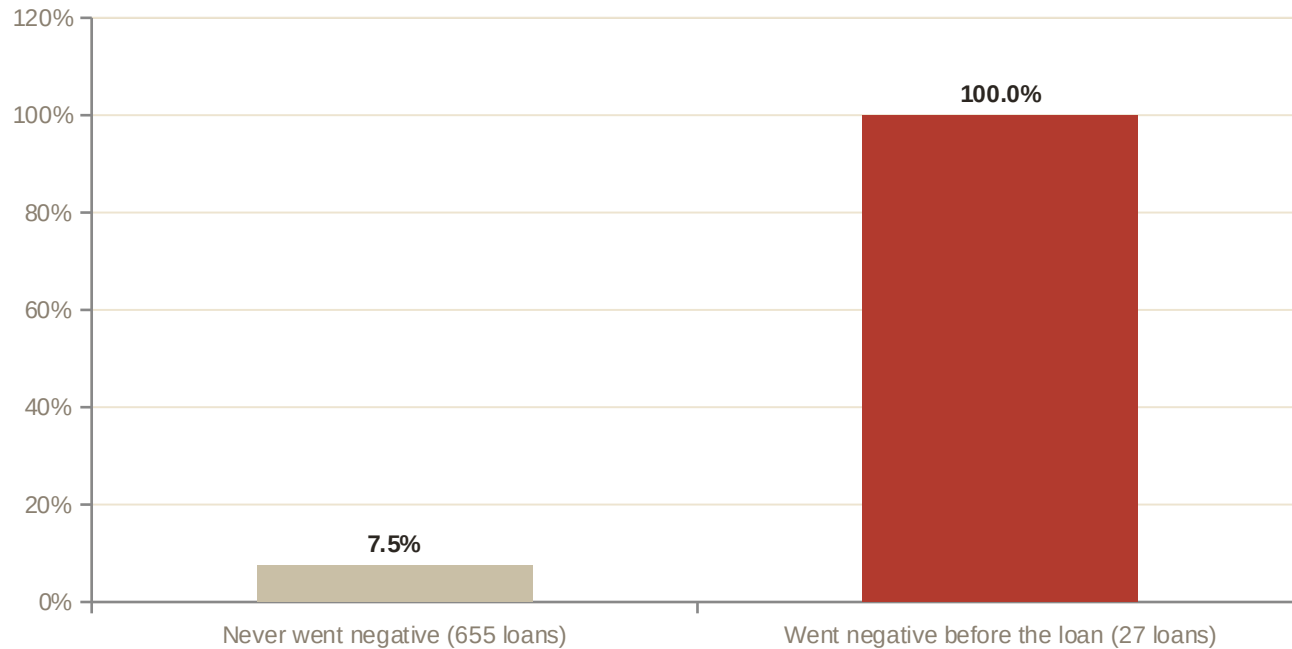
WHY IT MATTERS

Read as a real improvement, it would tell the bank its underwriting is already fixed. The honest baseline is the 13 to 14% of the mature cohorts.

Decision implication. The KPI must be reported by loan vintage at equal age, never as a single portfolio number. A dashboard that mixes cohorts will always make the newest lending look safest, which is precisely when standards get loosened by mistake. Loans originated per year: 20, 101, 90, 117, 196, 158.

Every pre-loan overdraft ended in a problematic loan

Problematic loan rate by pre-loan overdraft history, derived from the negative balances kept during cleaning.



WHAT HAPPENED

All 27 accounts that dipped below zero before applying became problematic, against 7.5% for everyone else. The flag alone accounts for 35.5% of every bad loan.

WHY IT HAPPENED

An account that cannot absorb its own everyday outflows has no buffer left to absorb a fixed monthly instalment on top of them.

WHY IT MATTERS

A single binary screen removes 27 of 76 problematic loans while touching only 4% of applicants.

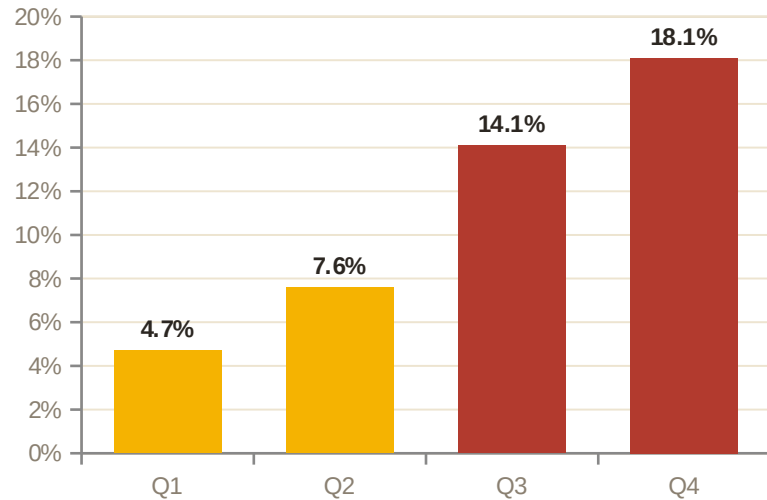
Honest caveats. The group is small at 27 loans, so 100% will not hold exactly on new data, and the flag is high precision but low recall, catching roughly a third of failures. It is a screen for manual review, not a standalone scorecard.

Affordability, not loan size, is the cleanest risk gradient

The three Milestone 1 cash-flow metrics, problematic loan rate by quartile, roughly 170 loans per quartile. r = correlation with failure.

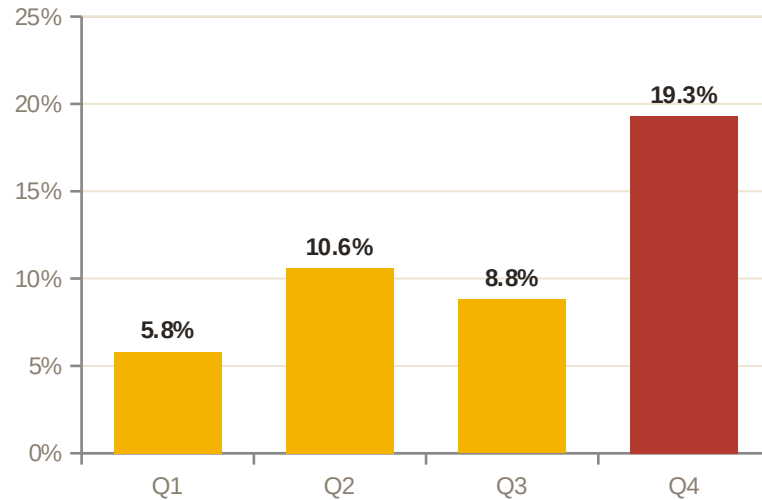
PAYMENT ÷ DISTRICT SALARY

monotonic 4.7 → 18.1 · $r = 0.18$



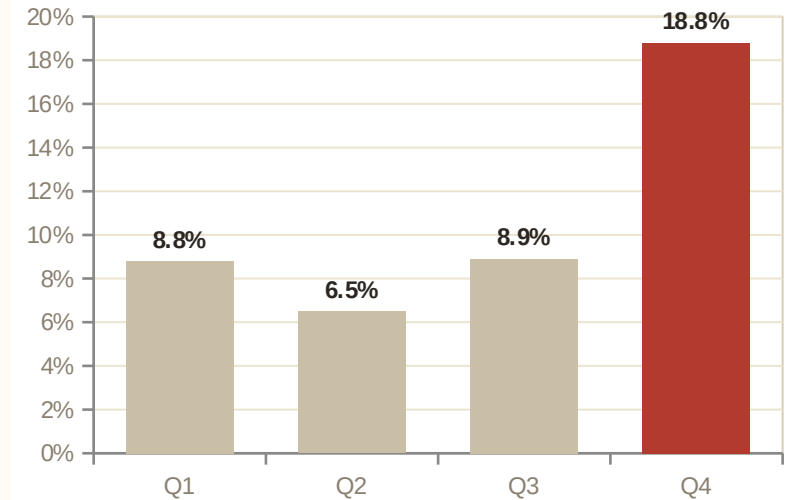
PAYMENT ÷ OWN MONTHLY INFLOW

strongest of the three · $r = 0.26$



OUTFLOW ÷ INFLOW

separates only at the extreme · $r = 0.04$

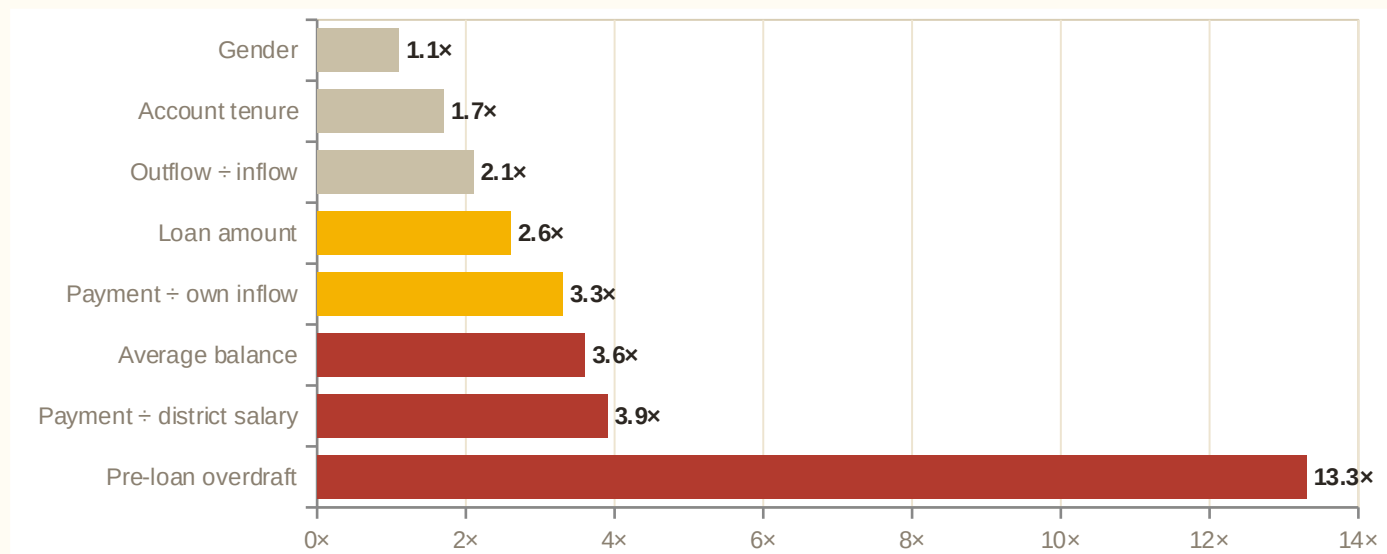
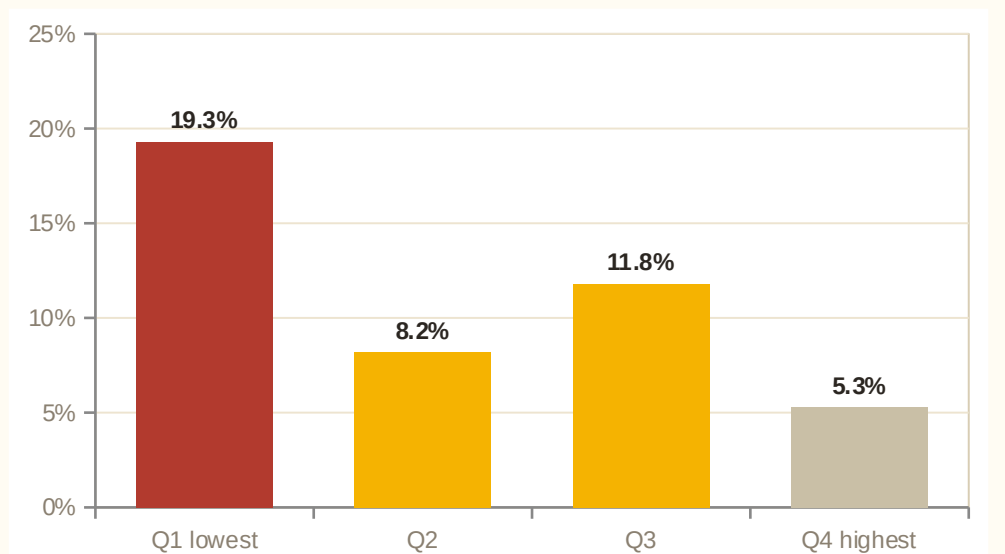


What happened: Payment-to-own-inflow shows the strongest risk gradient, rising from **5.8% to 19.3% ($r = 0.26$)**. Loan amount only increases risk in the top quartile (7.6% → 19.9%).

Why it matters: Repayment affordability matters more than loan size. Therefore, **payment-to-inflow should guide underwriting decisions** using existing application and account data.

The balance trough predicts risk, demographics do not

Left: problematic loan rate by average pre-loan balance quartile. Right: risk lift, the high-risk group rate ÷ the low-risk group rate, for the eight drivers tested.



THE TROUGH BEATS THE AVERAGE

Median average balance is Kč 44,113 for healthy loans against Kč 37,002 for problematic ones, but the minimum balance separates far better: Kč 700 against Kč 390. min_balance correlates with failure at -0.26 , avg_balance only at -0.16 . A customer can hold a comfortable mean and still hit zero every month before payday.

WHAT DOES NOT SEPARATE RISK

Gender shows little difference (11.8% female vs. 10.5% male), while age has no clear pattern. District unemployment is also negligible ($r = 0.018$). Behavioral indicators provide stronger, more direct risk signals than demographic or regional proxies.

All four root causes confirmed, four actions to take now

The Milestone 1 problem: the bank cannot tell before approval which applicants will end up in a problematic loan. Target: from 11% to below 5% within twelve months.

ROOT CAUSE · VERDICT · EVIDENCE

Weak account liquidity

CONFIRMED, strongest

100% failure rate after a pre-loan overdraft; -0.26 correlation on minimum balance

Loan burden

CONFIRMED, strongest positive

5.8% to 19.3% across quartiles, $r = 0.26$; loan amount only bites in its top quartile

Cash flow strain

CONFIRMED at the extreme

8.8% to 18.8%, but weak overall at $r = 0.04$ because the ratio is skewed

Thin banking history

WEAKLY CONFIRMED

15.2% for the newest tenure quartile against 8.8% for the longest, a secondary effect

FOUR ACTIONS THE CREDIT TEAM CAN TAKE NOW

01 Screen for a pre-loan overdraft

Route any applicant whose account went below zero in the last twelve months to manual review. Removes 27 of 76 problematic loans while touching 4% of applicants.

02 Cap payment-to-inflow at underwriting

Set a ceiling near the Q3 boundary so the instalment, not the principal, governs the decision. Strongest positive correlate at $r = 0.26$.

03 Add behavioural features to the scorecard

Feed min_balance, avg_balance and ever_negative into scoring. All three already sit in the warehouse.

04 Report the KPI by vintage, at equal age

Never publish one portfolio number. Compare each origination cohort at the same months-on-book, so the 1998 illusion cannot loosen standards.

What this analysis cannot yet claim, and how it was produced

Two layers. The consulting frame from Milestone 0 and 1 decides which questions are worth asking, CRISP-DM governs how they were answered.

LIMITATIONS, STATED OPENLY

Small event count

682 loans with only 76 problematic outcomes. The 27-account overdraft group is the most fragile finding, so its 100% rate needs revalidation on a larger sample.

Income is a proxy

Individual income is not in the dataset, so payment-to-income uses the district average salary. The ratio is directional rather than exact.

Association, not causation

Every result is a comparison of group rates. A low balance may be a symptom of the same pressure that causes default rather than its cause.

Period-bound data

1993 to 1998 in a transition-era Czech economy. Directions should generalise, exact thresholds should not be transplanted without recalibration.

WORKFLOW · CRISP-DM MAPPED TO THE MILESTONE STEPS

Business Understanding

Milestone 0 & 1

DARCI stakeholders, SMART problem statement, MECE issue tree, hypotheses and KPIs

Data Understanding

Step 1

Eight tables, keys, referential integrity at zero orphans, analysis unit 682 loans

Data Preparation

Step 2

Type conversion, missing values, duplicates, outliers, one row per loan under the pre-loan cut-off

Evaluation

Steps 3 to 5

Each Milestone 1 hypothesis tested, findings mapped back to the four root causes

Modeling & Deployment

Milestone 3

Scoped out on purpose: this milestone stops at association, the scoring layer and dashboard follow next

NOTEBOOK

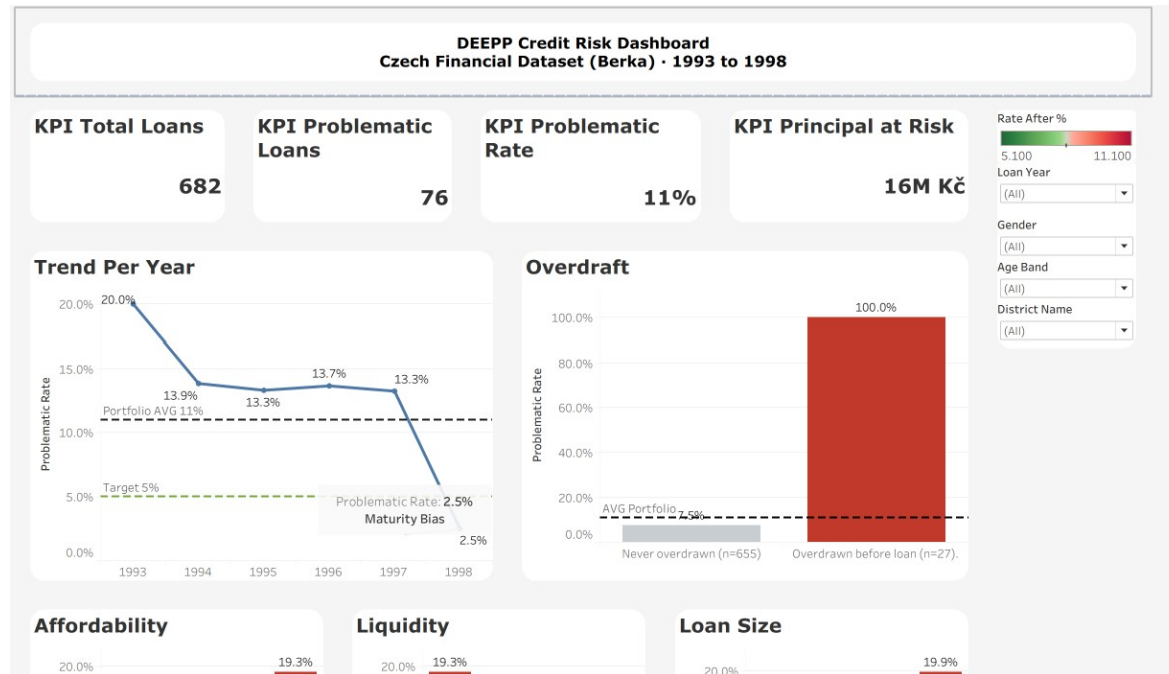
Full cleaning and analysis code, Python with DuckDB for the eight-table join, reproducible from the Kaggle source in one run.

Google Colab :

<https://colab.research.google.com/drive/1JrZ3ZMpgqfUzcoVN8JqiJjYGCVpZRgF2?usp=sharing>

One dashboard shows where the risk sits and what fixes it

Red marks the risk group only. Every bar starts at zero and is read against the dashed 11.1% portfolio average.



WHY EACH CHART TYPE WAS CHOSEN

KPI scorecard

four numbers at a glance

Line with average and target

direction over time, gap to 5%

Column, risk group in red

compares quartiles of each driver

Column by risk band

checks the model score ranks risk

Horizontal bar vs target

ranks each action by the rate left

TOOLS · WHY PYTHON AND TABLEAU

Python · Colab + DuckDB

Joins eight tables and 1.05M transactions, builds pre-loan features without leakage, runs the logistic regression and the impact simulation. Too heavy for a spreadsheet or Tableau.



loan_master_tableau.csv

One row per loan with the risk score and rule flags, the hand-off file.



Tableau Public

Interactive view for the credit team, filtered by year, gender, age band and district without opening any code

How to read each panel of the dashboard

Panel titles in Tableau are kept short. This guide gives the definition, the quartile order and the key insight behind each one.

PANEL	WHAT IT SHOWS	KEY INSIGHT	SLIDE
Trend Per Year	Rate by origination year. Dashed lines: 11.1% average, 5% target.	1998 at 2.5% is maturity bias, mature years sit at 13 to 14%.	16
Overdraft	Accounts that went below zero before the loan vs those that never did.	All 27 overdrawn accounts went bad, against 7.5% for the rest.	17
Affordability	Payment ÷ own monthly inflow. Q1 lowest burden, Q4 highest.	Q4 fails at 19.3%, Q1 only at 5.8%.	18
Liquidity	Average pre-loan balance. Q1 lowest balance, Q4 highest.	Q1 fails at 19.3%, Q4 only at 5.3%.	18
Loan Size	Loan amount. Q1 smallest loans, Q4 largest.	Flat at 7.6 to 9.4% until Q4 jumps to 19.9%.	18
Risk Band	Model score split into five equal groups, Very low to Very high.	Very high fails at 35.0% and holds 48 of 76 bad loans.	19
Simulation	Rate left after each screen. Grey vertical line marks the 5% target.	R1 + R3 is the only screen that reaches 5.0%.	20

SIMULATION VALUES

Current portfolio 11.1%

R2 Payment ÷ inflow > 0.17 8.4%

R1 Overdraft review 7.5%

R1 + R2 5.8%

R3 Risk score top 20% 5.1%

R1 + R3 5.0%

Colour scale. The Rate After % legend runs from green at 5.1% to red at 11.1%. The greener the bar, the closer the screen gets to the 5% target.

1 in 9 loans goes bad, more than twice the 5% target

Problematic Loan Rate = $(B + D) \div \text{total loans}$, by origination year, against the portfolio average and the Milestone 1 target.

682

loans analysed

76

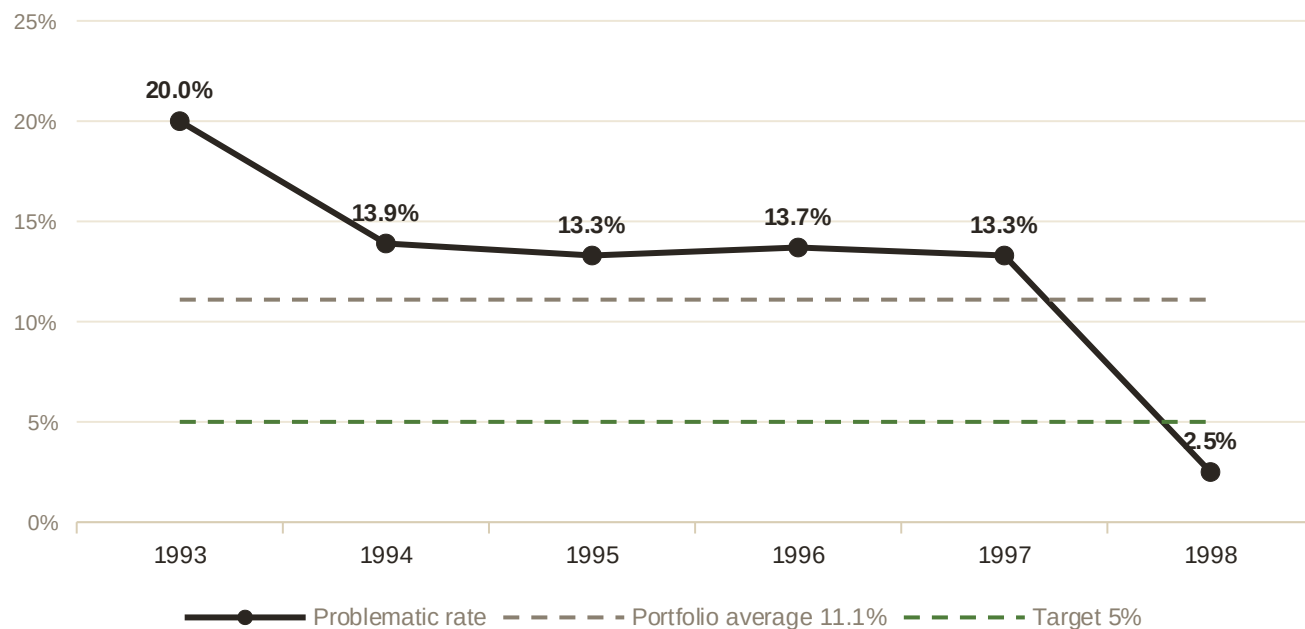
problematic loans (B + D)

11.1%

problematic rate

15.6M Kč

principal at risk



INSIGHT

Mature cohorts from 1994 to 1997 fail at 13 to 14%. The honest gap to the 5% target is 8 to 9 points, not 6.

BUSINESS INTERPRETATION

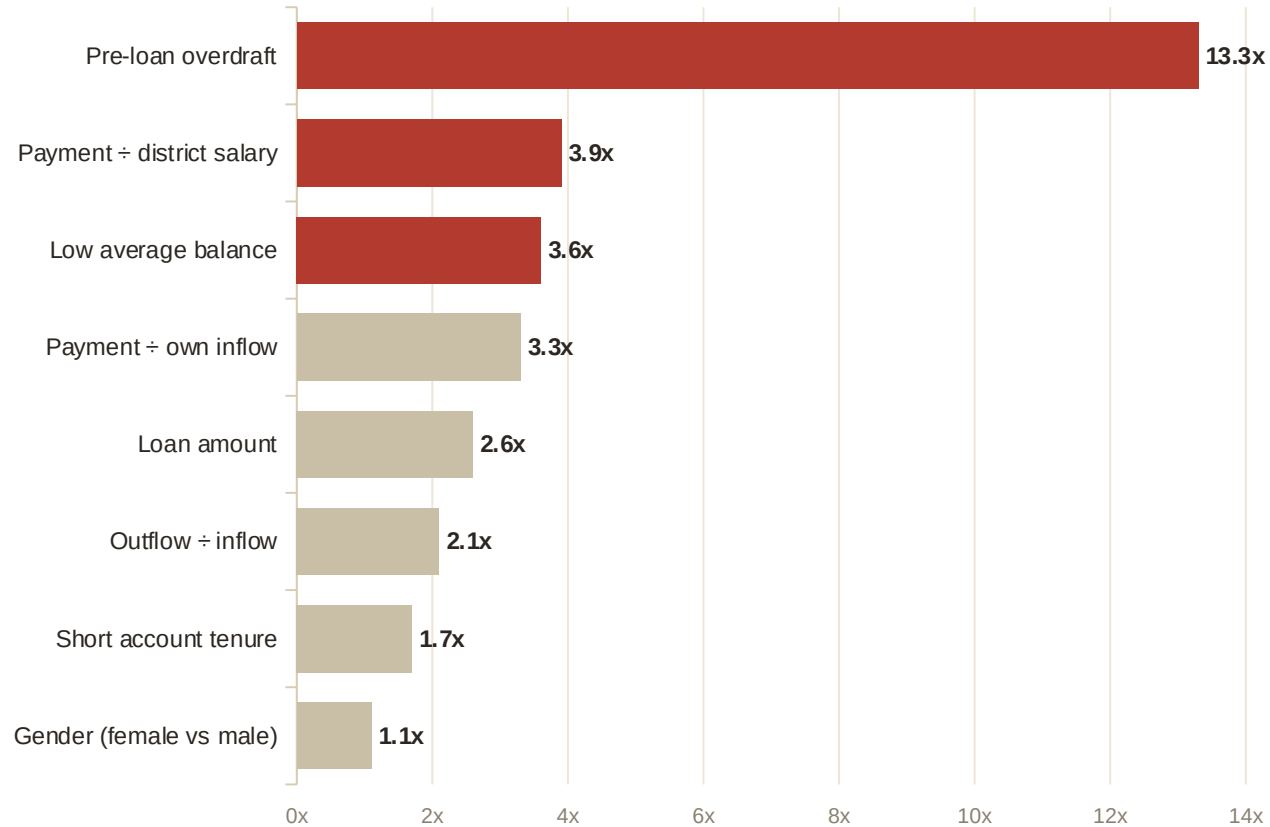
The 1998 value of 2.5% is maturity bias: those loans had under twelve months to go bad. Read as progress, it would loosen standards.

RECOMMENDATION

Track the KPI by vintage at equal months on book, and re-read the 1998 cohort after 24 months.

Account behaviour ranks far above who the customer is

Risk lift = problematic rate in the high-risk group ÷ rate in the low-risk group. 1.0x means no difference.



INSIGHT

A pre-loan overdraft multiplies risk 13.3 times. Affordability and a low balance follow at 3.6x to 3.9x. Gender sits at 1.1x.

BUSINESS INTERPRETATION

Risk shows up in how the account behaves, and the bank already records that behaviour before it approves the loan. Demographic screens would be both unfair and ineffective.

RECOMMENDATION

Build approval rules on the three red signals and keep gender and age out of the decision.

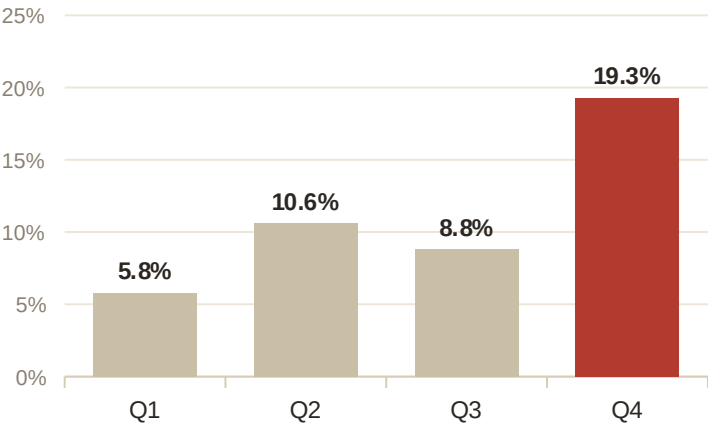
Top three drivers in red. Lift compares extreme quartiles, or the two groups for binary drivers.

Every driver's worst quartile fails at about 19%

Problematic loan rate by quartile, about 170 loans each. Portfolio average 11.1%. Risk quartile in red.

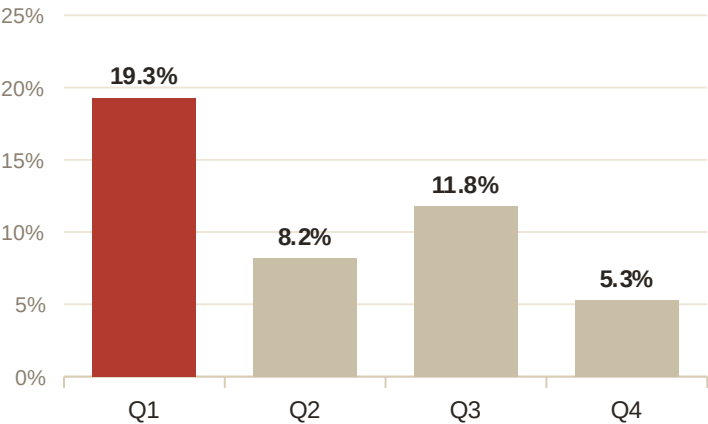
PAYMENT ÷ OWN MONTHLY INFLOW

Q1 lowest burden, Q4 highest



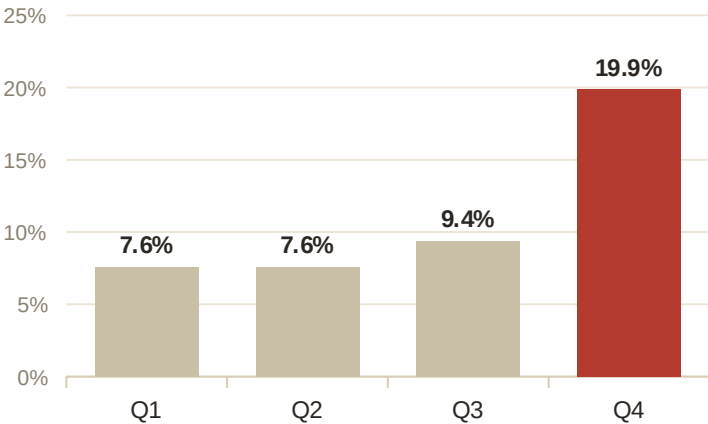
AVERAGE PRE-LOAN BALANCE

Q1 lowest balance, Q4 highest



LOAN AMOUNT

Q1 smallest loans, Q4 largest



INSIGHT

Stretched instalments and the lowest balances both fail at 19.3%, against 5 to 6% at the safe end. Loan amount stays flat at 7.6 to 9.4% until its top quartile.

BUSINESS INTERPRETATION

Large loans look risky mainly because their instalment eats the monthly inflow. Affordability is the lever, loan size is its symptom.

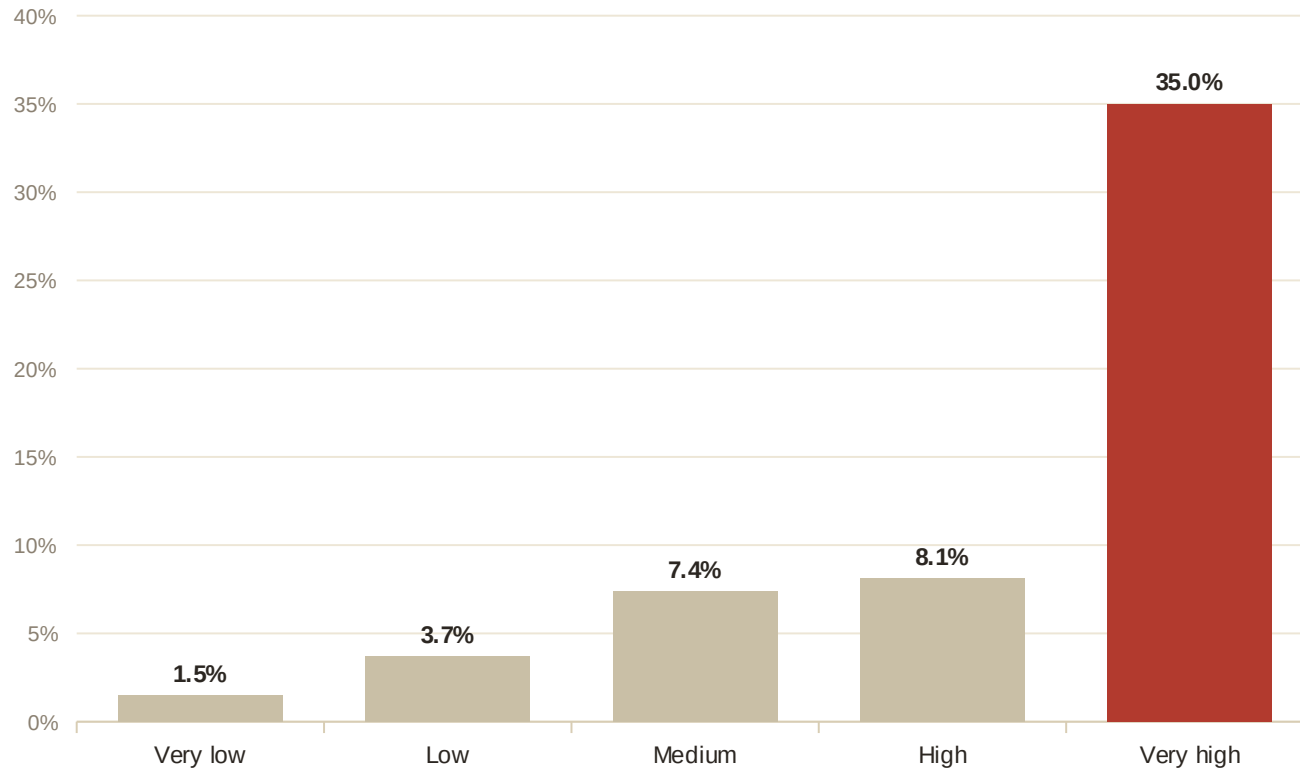
RECOMMENDATION

Cap payment ÷ inflow at 0.17, the start of Q4, rather than capping loan amount. Send bottom-quartile balances to a second review.

A simple risk score puts 48 of 76 bad loans in the riskiest 20%

Logistic regression on eight pre-loan features, 5-fold cross validation, so every loan is scored by a model that never saw it.

PROBLEMATIC LOAN RATE BY RISK BAND (QUINTILES OF THE SCORE)



INSIGHT

The very high band fails at 35.0%, three times the portfolio. The very low band fails at 1.5%. The score ranks risk out of sample.

BUSINESS INTERPRETATION

Features the bank already stores, balance, overdraft history and instalment burden, are enough to sort applicants before approval. 63% of bad loans sit in one fifth of applicants.

RECOMMENDATION

Route the top 20% of scores to manual review. Recalibrate on a larger sample before letting the score decline anyone automatically.

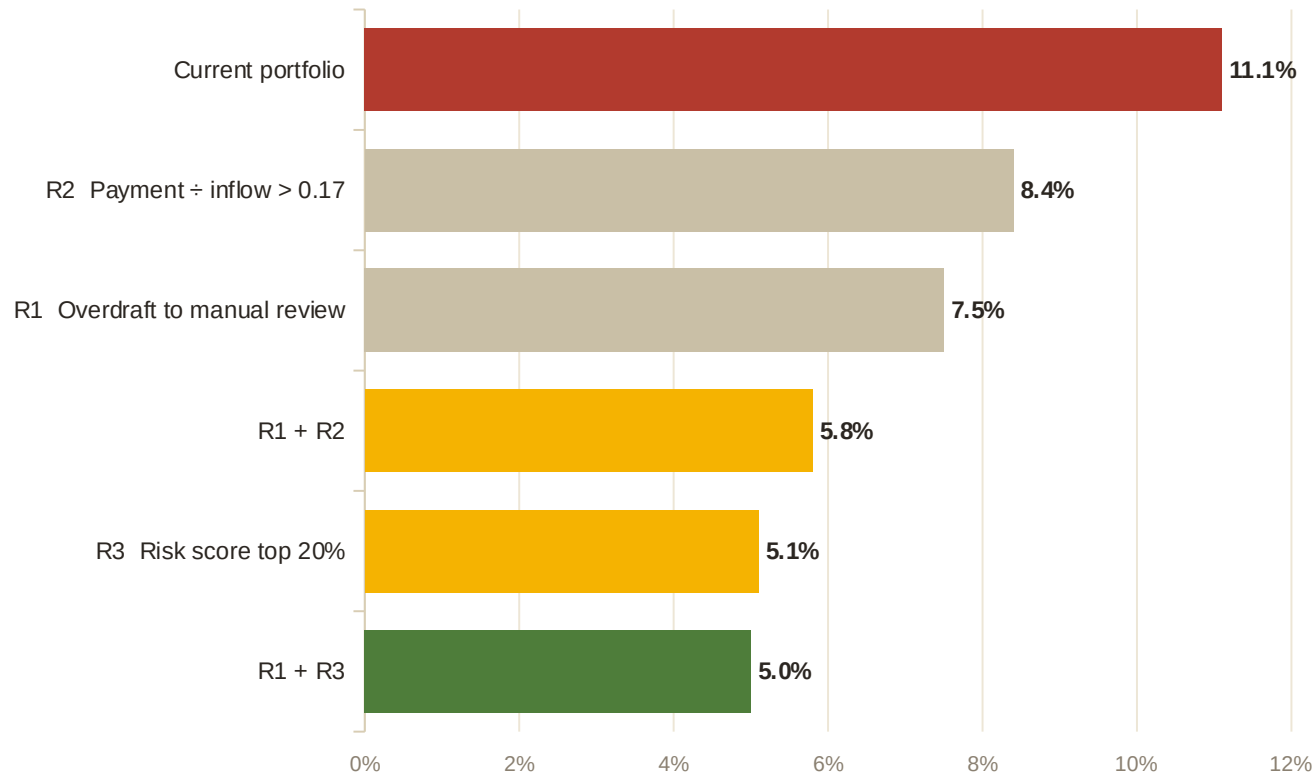
63%

of problematic loans captured by reviewing only the top 20% of scores

Overdraft screen plus risk score brings the rate to 5%

Problematic rate left in the portfolio after each screen. Assumption: a flagged application is declined or restructured.

RATE AFTER EACH SCREEN · TARGET 5%



INSIGHT

The overdraft screen alone cuts the rate to 7.5% and turns away no good loan. Adding the risk score review brings it to 5.0%.

BUSINESS INTERPRETATION

R1 + R3 flags about one in five applicants and removes 49 of 76 bad loans. The cost is roughly 89 good applicants sent to review, not necessarily declined.

RECOMMENDATION

Launch R1 now, run R3 as a review queue, and keep the R2 affordability cap as a buffer because 5.0% sits exactly on the target.

Colour key. Green reaches the target, amber lands within one point, grey is a partial fix, red is today.

Four findings, four actions, one path from 11.1% to 5%

Every action uses data available before disbursement and maps back to a Milestone 1 root cause.

FINDING	INSIGHT	ACTION	EXPECTED IMPACT
01 All 27 overdrawn accounts failed, 35.5% of bad loans	The strongest warning is already in the bank	Manual review for any overdraft in the last 12 months	11.1% → 7.5%, no good loan lost
02 Payment ÷ inflow: 5.8% in Q1, 19.3% in Q4	Affordability, not loan size, drives default	Cap payment ÷ inflow at 0.17 at underwriting	With R1: 5.8%
03 Riskiest 20% of scores hold 48 of 76 bad loans	Behaviour predicts risk, demographics do not	Add min and avg balance and overdraft flag to the scorecard, review the top 20%	With R1: 5.0%, on target
04 1998 at 2.5% vs 13 to 14% for mature cohorts	The recent improvement is maturity bias	Report the KPI by vintage at equal loan age	Prevents false progress

Next steps. Pilot R1 and R3 on new applications for three months, monitor the rate by vintage on the dashboard, and revalidate the score on a larger sample since 76 events limit precision.

Notebook: <https://colab.research.google.com/drive/1e9emuAy44EXOmNILx93IDxuazRG6gOMy?usp=sharing>

Dashboard: https://public.tableau.com/shared/9DCRP6QSK?:display_count=n&:origin=viz_share_link